

Abdelrhman Ahmed

Software Engineer | AI & Machine Learning Developer

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Military Service: Postponed | Willing to Travel: Yes

PROFESSIONAL SUMMARY

Computer Science student with hands-on experience in Python, AI, Machine Learning, Computer Vision, Full-Stack and Backend Development. Skilled in building scalable applications, RESTful APIs, pose estimation systems, and data analysis pipelines. Published co-author in a peer-reviewed Elsevier journal. Seeking a software engineering or AI/ML internship to apply skills in a professional environment.

EDUCATION

Bachelor of Science in Computer and Informatics

Oct 2023 - Jul 2027

Suez University, Suez, Egypt | Current GPA: 3.56 / 4.00 (Very Good) | Third year

WORK EXPERIENCE

Software Engineer (Part-Time) | CareCode Startup

Oct 2024 - Present

- Led development of Label-It, an AI support system for FLEXI App, integrating 12 exercise modules for real-time posture and fitness analysis
- Built backend services for generating and managing biomechanical analysis reports
- Developed a computer vision system using MediaPipe and Python to detect posture problems during exercises
- Designed scalable architecture to support efficient addition of new exercise modules

PROJECTS

Power Generation Forecasting using AI | Data Analysis & AI Development

2024 - 2025

- Performed exploratory data analysis and feature engineering to model solar PV power generation patterns
- Developed predictive forecasting models using Bayesian optimization and stacked ensemble learning to improve accuracy and robustness
- Optimized model hyperparameters and evaluated performance across multiple metrics
- Co-authored a peer-reviewed publication: "Photovoltaic Power Generation Forecasting with Bayesian Optimization and Stacked Ensemble Learning" Results in Engineering (Elsevier), 2025. DOI: <https://doi.org/10.1016/j.rineng.2025.104950>

Label-It — AI Support System for FLEXI App | Computer Vision & Backend Engineering

2024 - Present

- Built an end-to-end pose estimation pipeline to extract body keypoints, calculate biomechanical joint angles, and detect posture issues in real time
- Developed a state-machine-based exercise recognition system with threshold guards and motion validation for accurate movement tracking
- Implemented a rule-based posture classification engine and angle-based calculators for exercises such as squats and jumping jacks
- Integrated BLE connectivity to combine wearable sensor data with computer vision output, handling synchronization, drift, and reconnection
- Designed a scalable backend architecture for extending exercise modules efficiently

FLEXI — AI Coaching Application | Full-Stack AI & Backend Engineering

2024 - Present

- Developed a full-stack AI backend for automated posture analysis and flexibility assessment using pose detection
- Designed scalable RESTful APIs for analysis reporting and repetition tracking
- Built a high-performance PostgreSQL layer to store and query biomechanical metrics efficiently

- Created data pipelines to transform raw pose data into actionable fitness insights
- Implemented standardized error handling, automated testing with pytest, and optimized API performance

NoteNest — Collaborative Note-Taking Platform | Full-Stack Development

2024

- Designed and developed a collaborative web platform enabling students and teachers to create, organize, and share notes in secure virtual classrooms
- Architected the application using MVC design pattern ensuring clean separation of concerns and maintainability
- Engineered a database-driven backend with optimized MySQL schema design, migrations, and efficient query handling

TECHNICAL SKILLS

Programming Languages: Python, PHP, C++, JavaScript

AI & Machine Learning: TensorFlow, Scikit-Learn, Deep Learning, Computer Vision, Pose Estimation, MediaPipe

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Bayesian Optimization, Ensemble Learning

Web & Backend: Flask, FastAPI, RESTful APIs, MVC Architecture

Databases: PostgreSQL, MySQL, MongoDB, GraphQL

DevOps & Cloud: Docker, Docker Compose, AWS, Git, GitHub

Testing: pytest, Automated Testing

CERTIFICATIONS

- Machine Learning Specialization — DeepLearning.AI & Stanford University (Coursera)
- TensorFlow for Deep Learning Bootcamp — Zero To Mastery Academy
- Complete AI & Machine Learning, Data Science Bootcamp — Udemy
- Complete MLOps Bootcamp with 10+ End-to-End ML Projects — Udemy
- Data Structures & Algorithms — Zero To Mastery Academy
- SQL & Databases Bootcamp — Zero To Mastery Academy

ACHIEVEMENTS

- 1st Place — Data Science Track, Clash of Codes 2026
- 2nd Place — Hult Prize Egypt (Global Social Entrepreneurship Competition)
- GenZ Finalists — ISF (Innovation and Startup Festival)

PROFESSIONAL TRAINING

- Business Training — Creativa Ismailia (2026): Market Research, Team Building, Business Validation, Pitching & Presentation Skills
- Business Training — DenVC (2024): Startup Strategy, Entrepreneurial Skills

SOFT SKILLS

Problem Solving & Analytical Thinking | Communication & Team Collaboration | Leadership & Initiative | Adaptability & Continuous Learning | Time Management | Creativity & Innovation

LANGUAGES

Arabic (Native) | English (Professional Working Proficiency)